Arista Access Node YEL-ACC-32 Critical Failure SOP: Router Protocol Flap Resolution

Node: YEL-ACC-32 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_1914

Description

In the Arista network architecture, the Access Node YEL-ACC-32 at the Yelahanka Sub-station is experiencing a critical failure due to a Router Protocol Flap, manifesting as a BGP route churn, OSPF adjacency loss, and ISIS LSP flooding. This SOP aims to identify the root cause of the failure, diagnose the issue, and implement a resolution. The Access Node's role in the network involves establishing and maintaining routing protocol sessions with upstream and downstream nodes. However, in cases where the routing protocol sessions fail or experience high churn, it can lead to route instability, packet loss, and network congestion. On the Arista EOS platform, the CLI dialect plays a crucial role in diagnosing and resolving such issues. As the Access Node is equipped with Arista's proprietary EOS software, the subsequent steps are tailored to utilize the EOS CLI to troubleshoot and resolve the Router Protocol Flap.

Pre-requisite Checks

- Check: Verify the power supply and cooling systems at the Yelahanka Sub-station are functioning correctly and within normal operating ranges.

- Check: Validate the redundant link status between the Access Node YEL-ACC-32 and its upstream/downstream nodes.

- Check: Confirm that all relevant alarms and logs have been collected and reviewed for any recent events or issues.

Potential Root Causes

Cause	Probability
BGP Route Server Failure	High
OSPF Neighbour Failure	Medium
ISIS LSP Flooding due to Loop Detection	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP table version is 0, main routing table version is 0, Description: This command is used to verify the BGP routing table and its associated statistics. It helps in identifying any issues related to BGP route churn or neighbour failure.

Command: show ip ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: This command is used to verify the OSPF neighbour status and its associated statistics. It
```

```
helps in identifying any issues related to OSPF adjacency loss or
neighbour failure.

Command: show isis database, Expected Output: ISIS LSP Database,
Description: This command is used to verify the ISIS LSP database and
its associated statistics. It helps in identifying any issues related
to ISIS LSP flooding or loop detection.
```

Resolution Steps

Step 1: Step: Reboot the BGP Route Server, Command: reload, Description: Rebooting the BGP Route Server can sometimes resolve issues related to BGP route churn or neighbour failure. This step is essential to ensure that the BGP Route Server is functioning correctly.

Step 2: Step: Reset OSPF Neighbours, Command: clear ip ospf neighbor, Description: Resetting OSPF neighbours can sometimes resolve issues related to OSPF adjacency loss or neighbour failure. This step is essential to ensure that the OSPF neighbours are functioning correctly.

Step 3: Step: Clear ISIS LSP Database, Command: clear isis database, Description: Clearing the ISIS LSP database can sometimes resolve issues related to ISIS LSP flooding or loop detection. This step is essential to ensure that the ISIS LSP database is functioning correctly.

Post-Check Verification

```
Command: show bgp summary, Expected Output: BGP table version is 0,
main routing table version is 0, Description: This command is used to
verify the BGP routing table and its associated statistics after
resolving the issue.

Command: show ip ospf neighbor, Expected Output: Neighbor ID Pri State
Dead Time Address Interface, Description: This command is used to
verify the OSPF neighbour status and its associated statistics after
resolving the issue.

Command: show isis database, Expected Output: ISIS LSP Database,
Description: This command is used to verify the ISIS LSP database and
its associated statistics after resolving the issue.
```

Roll-back Procedures

- Step: Restore the Original Configuration, Command: load -b /running-config, Description: In case the resolution steps fail, restoring the original configuration can help revert the node to its previous state. This step is essential to prevent any further disruptions to the network.